

Skills Summary

Expanded Form, Comparing, and Rounding

Use this reference while completing your worksheet or when studying.

1. Expanded form: what each digit is worth

Every place is ten times the one to its right.

Ten thousands	Thousands	Hundreds	Tens	Ones
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Expanded form writes the number as the sum of those values. A place with no digit still gets a zero in standard form — the placeholder is what keeps every other digit where it belongs.

Example: Write 4907 in expanded form.

Step 1. Expanded form asks for each digit's value, so name the place of every digit before writing anything.

Step 2. 4 thousands, 9 hundreds, 0 tens, 7 ones, so the values are 4000, 900, nothing, and 7.

$$\Rightarrow 4907 = 4000 + 900 + 7$$

Try it: Write 5263 in expanded form.

$$5000 + 200 + 60 + 3 = 5263 \text{ check}$$

2. The value of one digit

Digit or value? The *digit* is the symbol you see; the *value* is the digit times its place. In 3600 the digit 6 has the value 600.

Same digit, two places: each step to the left multiplies the value by 10. Two places apart is 100 times; three places apart is 1000 times.

Example: In 2640, what is the value of the 6?

Step 1. The question asks for a value, not a digit, so find which place the 6 sits in and multiply.

Step 2. Counting from the right — ones, tens, hundreds — the 6 is in the hundreds place, so its value is 6×100 .

$$\Rightarrow 600$$

Try it: In 19342, what is the value of the 9?

$$90000 \text{ check}$$

3. Comparing: biggest place, not biggest digit

Rule: line the numbers up by place and start at the *left*. Move right only while the digits are equal. The first place where they differ decides the whole comparison — nothing further right can change it.

The small point of $<$ and $>$ aims at the smaller number.

Example: Write $<$ or $>$ between 52861 and 52816.

Step 1. Comparing starts at the largest place, so walk left to right until the digits stop matching.

Step 2. Ten-thousands, thousands and hundreds all match (5, 2, 8). The tens place is the first difference: 6 against 1.

$\Rightarrow$ **52861 $>$ 52816**

Try it: Write $<$ or $>$ between 30472 and 30742.

check: $30472 < 30742$

4. Rounding to a named place

Three steps: find the place you are rounding to; look at the digit *one place to its right*; five or more rounds up, four or less stays. Then every digit to the right becomes zero.

Watch the carry: if the rounding place holds a 9 and it rounds up, the extra one moves into the next place left — ten hundreds is one thousand.

Example: Round 6473 to the nearest hundred.

Step 1. Rounding to hundreds is a choice between two hundreds, so the digit that decides is the one just to the right — the tens digit.

Step 2. The tens digit is 7, which is five or more, so the hundreds digit goes from 4 up to 5 and the rest become zero.

$\Rightarrow$ **6500**

Try it: Round 24681 to the nearest thousand.

check: 25000